

Muhammad Munhib Baig

Karachi, Pakistan | munhibbaig@gmail.com | +92 318 2109157 | [LinkedIn](#) | [GitHub](#)

Professional Summary

AI and Machine Learning Engineer with a Computer Science degree from FAST-NUCES and practical experience building deep-learning-based computer vision systems, federated learning pipelines, and LLM-integrated AI agents. Skilled in PyTorch, YOLOv8, OpenCV, and Scikit-learn; experienced integrating ML models into production backends and agentic workflows using Python.

Experience

Codify Labs

Software Developer Intern

July 2026 – Present

Lahore, Pakistan

- Engineered an OCR pipeline integrating the **Anthropic API** to extract handwritten student answers from exam images; implemented concurrent processing and structured error handling, reducing manual grading effort significantly.
- Built a PDF extraction engine using **PyMuPDF**, **OpenCV**, and **NumPy** to detect and isolate answer regions across exam papers, eliminating false detections through duplicate suppression and region filtering.
- Delivered end-to-end UI components unifying document upload, OCR, AI grading, review, and result export into a single seamless workflow, applying modular design and secure environment configuration throughout.
- Proposed and implemented workflow improvements across the document extraction, AI evaluation, and result-generation pipeline, accelerating development delivery and improving system cohesion.

Education

National University of Computer and Emerging Sciences (FAST-NUCES)

Bachelor of Science in Computer Science

Aug 2022 – Jun 2026

Relevant Coursework: Software Engineering, Web Technologies, Database Systems, Operating Systems, Computer Networks, Cloud Computing, Artificial Intelligence, Machine Learning, Data Science

Projects

A.R.E.S — Autonomous Response & Emergency Surveillance (FYP)

AI/ML Engineer — Final Year Project

2025–2026

Python, YOLOv8, OpenCV, Flower, FastAPI

- Designed a privacy-preserving AI surveillance system detecting criminal activities—**theft, fighting, vandalism, weapon detection**—across distributed CCTV streams using **Federated Learning (Flower)**.
- Improved detection model accuracy from **73.8% to 83.9%** by aggregating locally trained **YOLOv8** models across four simulated edge clients without transferring raw footage.
- Built a real-time inference pipeline with **OpenCV** achieving **sub-two-second** alert generation latency, served through a **FastAPI** backend to an Electron desktop interface.
- Applied federated aggregation strategies (FedAvg) to maintain model performance while ensuring data residency compliance on edge devices.

MentorOS — AI Agent Platform

AI Integration Engineer — Collaborative Project

2026

Python, MCP, LLMs

- Contributed to an LLM-powered software engineering assistant leveraging **AI agents** for automated code generation, project understanding, and development workflow assistance.
- Implemented **Model Context Protocol (MCP)** integrations and agent orchestration modules enabling multi-step reasoning and tool-use across configurable LLM providers.
- Designed memory-handling and configuration management components supporting model-agnostic operation across multiple LLM backends (Ollama, OpenAI, Anthropic, OpenRouter).

Business Performance Dashboard

Data Analyst — Personal Project

2024

Power BI, Tableau, Excel

- Analysed a retail sales dataset of **10,000+ records** to surface revenue trends, regional performance, and product-level insights using Power BI and Tableau.
- Created interactive dashboards enabling stakeholders to identify high-performing SKUs and monitor KPIs for data-driven decisions.